
Math Training — M1 Cognitive Science

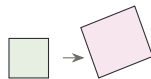
Session 2 — Linear Algebra II

Determinant, rank, eigenvectors, diagonalisation, PCA, SVD

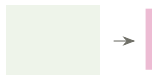
Advanced track

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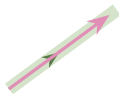
1. **Determinant**, inverse, **rank**
2. **Low rank**: a big table acting small
3. **Eigenvectors**, eigenvalues, diagonalization
4. Symmetric matrices, covariance, **PCA**
5. **SVD**: PCA straight from the table



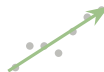
area factor



collapse

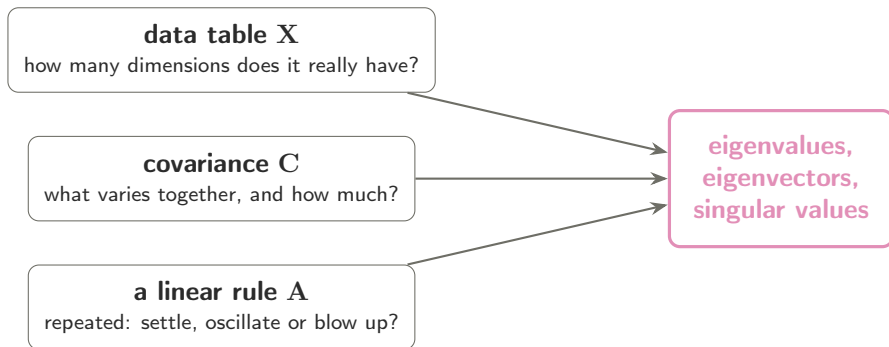


pure stretch



long axis

Three questions that all reduce to the spectrum

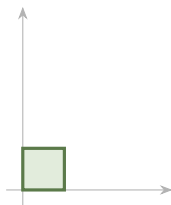


■ Key idea — one tool, three questions

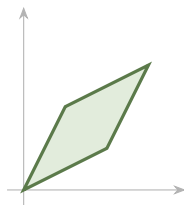
Which matrix you look at depends on the question. The answer is always its spectrum.

■ Definition — determinant of a 2×2 matrix

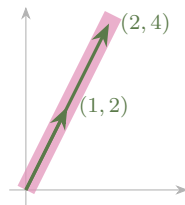
$$\det \mathbf{A} = a_{11}a_{22} - a_{12}a_{21}$$



area 1

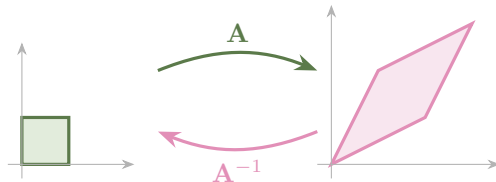


$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: area 3



$\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$: area 0

$|\det|$ multiplies every area; its *sign* says whether orientation is kept, so a reflection has a negative determinant. For $\mathbf{B}$ the two columns lie on one line: the parallelogram is flat.

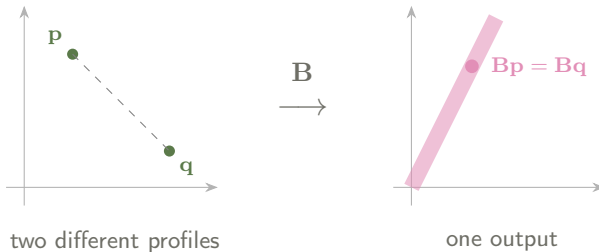


- **Definition — inverse of a matrix A :** the map that undoes it, $A^{-1}A = I$

When $\det A \neq 0$,

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

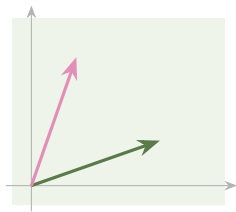
— swap the diagonal, negate the off-diagonal, divide by $\det$.



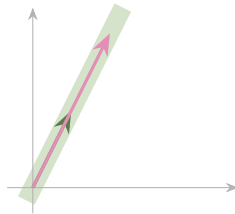
■ Why $\det = 0$ means “no inverse”

The map flattens the plane onto a line, so two different inputs give the *same* output. There is nothing to invert: the information is lost, not hidden.

Rank: the dimension of the set of outputs



rank 2: outputs fill the plane

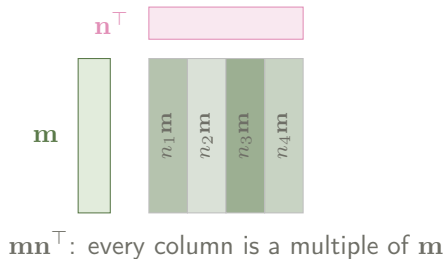


rank 1: outputs on one line

■ Definition — rank of a matrix

Number of independent columns = dimension of the set of outputs (the column reading of Session 1).

Rank 1 at any size: the outer product



■ Rank of an outer product

For $\mathbf{m}, \mathbf{n} \neq \mathbf{0}$, the matrix $\mathbf{m}\mathbf{n}^\top$ has **rank 1** whatever its size. A sum of r outer products has rank $\leq r$.

■ Example

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

$$\det \mathbf{A} = 4 - 1 = 3 \neq 0$$

$\Rightarrow$ invertible, **rank 2**

$$\mathbf{A}^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{A} \cdot \frac{1}{3} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \checkmark$$

■ Example

$$\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

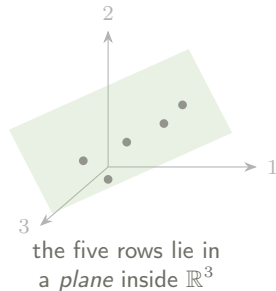
$$\det \mathbf{B} = 4 - 4 = 0$$

$\Rightarrow$ no inverse, **rank 1**

Outputs on the line through $(1, 2)$, and
 $\mathbf{B} = \mathbf{m}\mathbf{n}^\top$ with $\mathbf{m} = \mathbf{n} = (1, 2)$.

Rank in practice: a redundant column is collinearity

	item 1	item 2	total
P_1	4	9	13
P_2	6	7	13
P_3	5	8	13
P_4	9	4	13
P_5	7	6	13



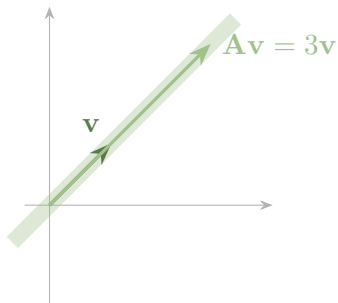
column 3 = column 1 + column 2
 $\Rightarrow$ **rank 2**, not 3

■ In a regression

A column built from the others adds nothing, so **many weight vectors give identical predictions**: “the effect of item 1” is not defined. This is **collinearity**.

■ Practice — try it

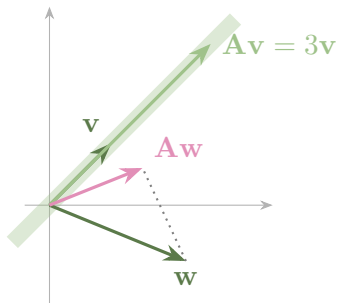
1. Compute $\det\begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$ and its inverse.
2. Rank of $\begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix}$? Invertible? On what line do its outputs lie?
3. For which k is $\begin{pmatrix} 1 & 2 \\ 2 & k \end{pmatrix}$ non-invertible? What happens to the rank there?



■ Definition — eigenvector and eigenvalue of A

$v \neq 0$ is an **eigenvector** if $Av = \lambda v$. The stretch factor λ is the **eigenvalue**.

v stays on the green line, three times longer; w is knocked off its own line.



■ Definition — eigenvector and eigenvalue of A

$\mathbf{v} \neq \mathbf{0}$ is an **eigenvector** if $A\mathbf{v} = \lambda\mathbf{v}$. The stretch factor λ is the **eigenvalue**.

$\mathbf{v}$ stays on the green line, three times longer; $\mathbf{w}$ is knocked off its own line.

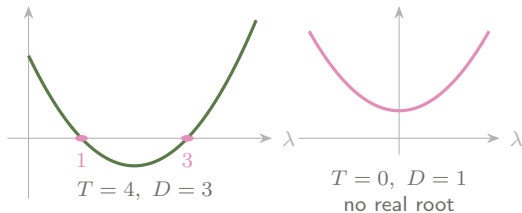
$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

$$\lambda^2 - T\lambda + D = 0$$

$$T = \operatorname{tr} \mathbf{A}, \quad D = \det \mathbf{A}$$

■ Two free checks

$$\lambda_1 + \lambda_2 = T, \quad \lambda_1 \lambda_2 = D.$$



Roots of the parabola. No crossing $\Rightarrow$ complex pair.

■ Example

$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: $T = 4, D = 3 \Rightarrow \lambda^2 - 4\lambda + 3 = 0 \Rightarrow \lambda = 1, 3$. Check: $1 + 3 = 4$
 $\checkmark, 1 \cdot 3 = 3 \checkmark$

$\lambda = 3$:

$$(\mathbf{A} - 3\mathbf{I})\mathbf{v} = \mathbf{0} \Rightarrow -v_1 + v_2 = 0$$

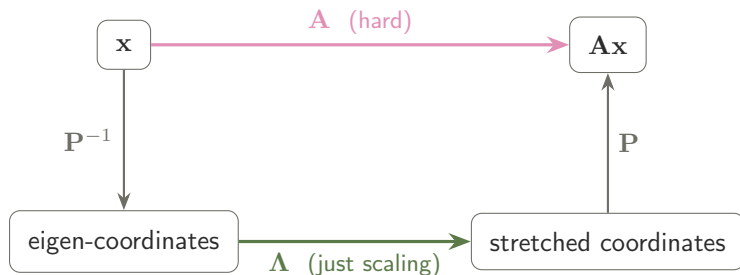
$$\mathbf{v} = (1, 1)$$

$\lambda = 1$:

$$(\mathbf{A} - \mathbf{I})\mathbf{v} = \mathbf{0} \Rightarrow v_1 + v_2 = 0$$

$$\mathbf{v} = (1, -1)$$

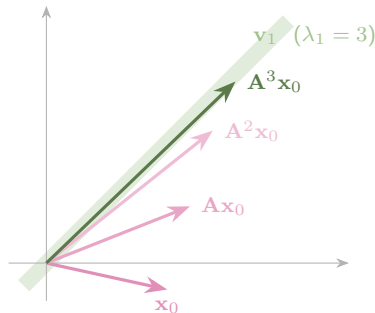
$(1, 1) \cdot (1, -1) = 0$: orthogonal. Not an accident — see the spectral theorem.



■ Diagonalization — and what “same order” means

Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be eigenvectors forming a *basis*, $\mathbf{A}\mathbf{v}_i = \lambda_i\mathbf{v}_i$. Put $\mathbf{v}_i$ in *column* i of $\mathbf{P}$ and λ_i in *diagonal entry* i of $\mathbf{\Lambda}$ — same index i in both. Then, column by column,

$$\mathbf{A}\mathbf{P} = (\lambda_1\mathbf{v}_1 \ \cdots \ \lambda_n\mathbf{v}_n) = \mathbf{P}\mathbf{\Lambda}, \quad \text{so} \quad \mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}, \quad \mathbf{A}^k = \mathbf{P}\mathbf{\Lambda}^k\mathbf{P}^{-1}.$$



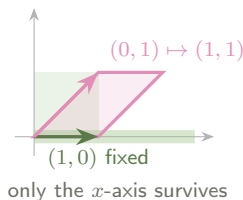
■ Key idea — the dominant mode takes over

In an eigenbasis, $\mathbf{A}^k \mathbf{x}_0 = c_1 \lambda_1^k \mathbf{v}_1 + c_2 \lambda_2^k \mathbf{v}_2$, and $(\lambda_2/\lambda_1)^k \rightarrow 0$: the direction converges to $\mathbf{v}_1$.

Provided $|\lambda_1|$ is *strictly* largest and $c_1 \neq 0$ — true for a random start.

■ Watch out

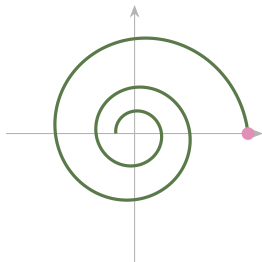
$\mathbf{S} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has $\lambda = 1$ *twice* but only *one* eigendirection, $(1, 0)$: **defective** — no eigenbasis, no diagonalization.



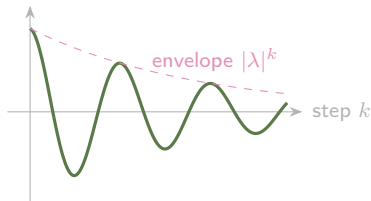
■ The exact condition

$\mathbf{A}$ ($n \times n$) is diagonalizable $\iff$ there are n *independent* eigenvectors $\iff$ for *every* λ , the number of independent solutions of $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ equals the multiplicity of λ as a root of $\det(\mathbf{A} - \lambda\mathbf{I})$.

Distinct eigenvalues are **sufficient**,
never necessary ($\mathbf{I}$ repeats $\lambda = 1$ and is diagonal).



$|\lambda| < 1$: spiral inward



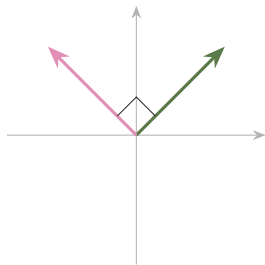
one coordinate of $\mathbf{x}_k$, step after step

■ Key idea — complex λ means turning

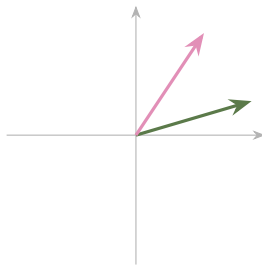
For a real matrix, complex eigenvalues come in pairs $\lambda, \bar{\lambda}$. Applying $\mathbf{A}$ once more, $\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k$, **turns the vector by $\arg \lambda$ and rescales it by $|\lambda|$** : rotation, with decay if $|\lambda| < 1$ and growth if $|\lambda| > 1$.

■ Practice — try it

1. Verify $(1, 0)$ is an eigenvector of $\begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}$; give its eigenvalue.
2. For $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ compute T and D , then the eigenvalues.
3. A matrix has eigenvalues 2 and 0.5. Apply it many times to a generic vector: which direction does the result line up with, and why?



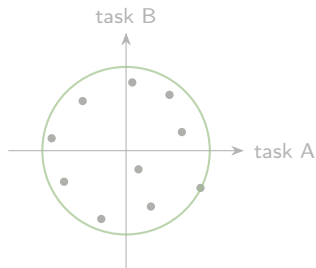
$\mathbf{A} = \mathbf{A}^T$
eigenvectors **perpendicular**



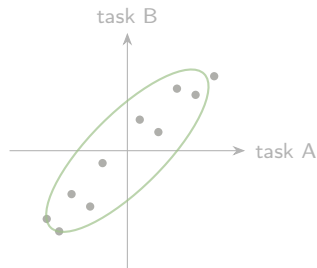
$\mathbf{A} \neq \mathbf{A}^T$
eigenvectors **oblique**

■ Spectral theorem

A *real symmetric* matrix has *real* eigenvalues and admits an *orthonormal basis of eigenvectors*. Collect that basis in $\mathbf{P}$: then $\mathbf{P}^{-1} = \mathbf{P}^T$ and $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^T$.



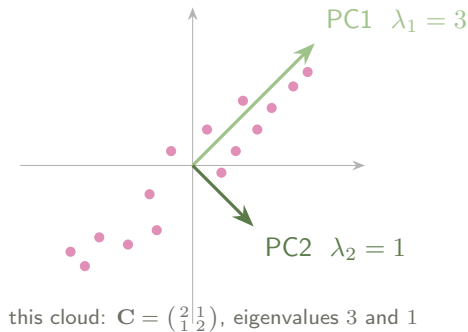
$\begin{pmatrix} v & 0 \\ 0 & v \end{pmatrix}$: uncorrelated



$\begin{pmatrix} v & c \\ c & v \end{pmatrix}$: correlated, tilted

■ Definition — covariance matrix $\mathbf{C}$ of a centred table $\mathbf{X}$

$\mathbf{C} = \frac{1}{n} \mathbf{X}^\top \mathbf{X}$: variances on the diagonal, co-variation off it. Symmetric, and **positive semi-definite** — for a *unit* $\mathbf{u}$, $\mathbf{u}^\top \mathbf{C} \mathbf{u}$ is the variance of the data projected on $\mathbf{u}$, hence ≥ 0 .



■ Key idea — principal components and their variances

New perpendicular axes, ordered by how much the data varies along each. Principal components = **eigenvectors of C** ; each eigenvalue = **variance along that direction**. Those variances add up to the total, $\sum_k \lambda_k = \text{tr } C$, so PC1's share is

■ Example

Two similar tasks, centred scores: $\mathbf{C} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

$\lambda_1 = 3$ along $(1, 1)$: **overall level**

$\lambda_2 = 1$ along $(1, -1)$: **B – A**

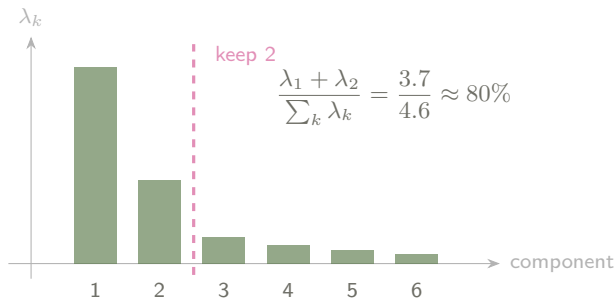
Both real, both ≥ 0 , and orthogonal — exactly as the spectral theorem and positive semi-definiteness promise.

$$\frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{3}{4} = 75\%$$

along $(1, 1)$; the remaining 25% along $(1, -1)$.

The rotation we *guessed* in Session 1 is the one PCA *finds*.

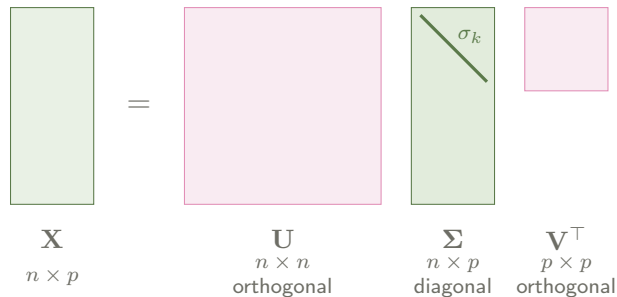
How many components?



■ Choosing how many to keep

Six tests, but the scores really move along *two* directions. Two honest criteria: the **cumulative variance explained**, and the **elbow**. Neither is a proof — this is a judgement, and should be reported as one.

The singular value decomposition

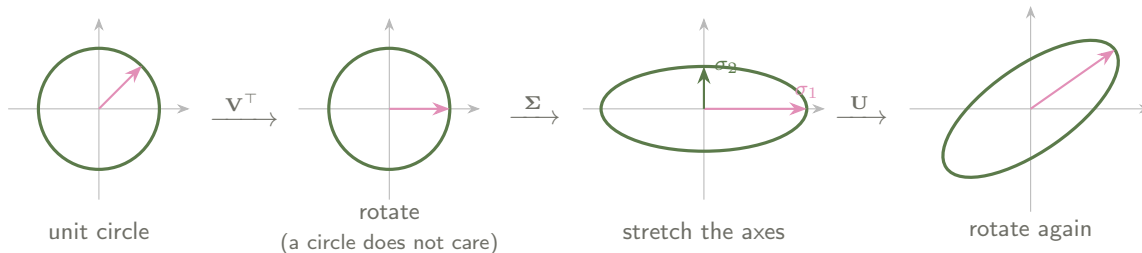


■ Definition — singular value decomposition of $\mathbf{X}$

$$\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$$

$\mathbf{U}, \mathbf{V}$ orthogonal ($\mathbf{U}^T \mathbf{U} = \mathbf{I}$, $\mathbf{V}^T \mathbf{V} = \mathbf{I}$); $\mathbf{\Sigma}$ diagonal, $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$. **Every** matrix factors this way — any shape, any rank, no exceptions.

Geometry of the SVD: rotate, stretch, rotate



■ Key idea — circle in, ellipse out

Every linear map is a **rotation**, a **stretch along perpendicular axes**, and **another rotation**. The σ_k are the semi-axes of the ellipse: how far the map reaches in each direction.

$$\mathbf{X}^\top \mathbf{X} = (\mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top)^\top (\mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top) = \mathbf{V} \mathbf{\Sigma}^\top \underbrace{\mathbf{U}^\top \mathbf{U}}_{=\mathbf{I}} \mathbf{\Sigma} \mathbf{V}^\top = \mathbf{V} (\mathbf{\Sigma}^\top \mathbf{\Sigma}) \mathbf{V}^\top$$

■ The two decompositions match

$\mathbf{\Sigma}^\top \mathbf{\Sigma}$ is diagonal with entries σ_k^2 , so the line above *is* an eigendecomposition. With $\mathbf{C} = \frac{1}{n} \mathbf{X}^\top \mathbf{X}$,

$$\underbrace{\text{columns of } \mathbf{V}}_{\text{right singular vectors}} = \underbrace{\text{eigenvectors of } \mathbf{C}}_{\text{principal components}}, \quad \lambda_k = \frac{\sigma_k^2}{n}.$$

■ Definition — variance explained by component k

Its share of the total variance: $\lambda_k / \sum_j \lambda_j = \sigma_k^2 / \sum_j \sigma_j^2$ (the n 's cancel).

PCA is the SVD of the centred data table.

$$\mathbf{X} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^\top + \cdots + \sigma_r \mathbf{u}_r \mathbf{v}_r^\top$$

keep these throw the rest away

■ Eckart–Young

Among *all* matrices of rank at most r , the truncation $\mathbf{X}_r = \sum_{k \leq r} \sigma_k \mathbf{u}_k \mathbf{v}_k^\top$ is the closest one to $\mathbf{X}$ in the least-squares (Frobenius) sense: it minimises $\sum_{ij} (X_{ij} - (X_r)_{ij})^2$.

■ What truncation is for

The table is a **sum of outer products**, biggest first — each term the rank-1 “one profile, one level per person” pattern from earlier. Keeping two turns twenty items into two component scores; the terms you drop, the small σ_k , are where the noise was living.

■ Example

Four participants, two measures, already centred: $\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ -1 & -1 \\ -1 & -1 \end{pmatrix}$

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix} : \quad T = 8, \quad D = 0$$

$$\Rightarrow \sigma_1^2 = 8, \quad \sigma_2^2 = 0 \Rightarrow \sigma_1 = 2\sqrt{2}, \quad \sigma_2 = 0$$

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1), \quad \mathbf{u}_1 = \frac{\mathbf{X}\mathbf{v}_1}{\sigma_1} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}$$

Rank 1: $\mathbf{X} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top$ exactly.

Variance explained by PC1:

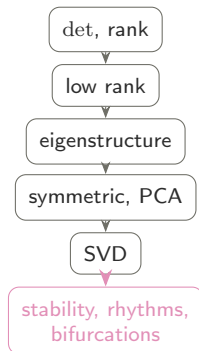
$$\frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} = \frac{8}{8} = 100\%.$$

One profile ($\mathbf{v}_1$: both measures together)
and one number per participant reproduce
the whole table.

■ Practice — try it

1. $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$: symmetric? Find its eigenvalues and check the eigenvectors are orthogonal.
2. A covariance matrix has eigenvalues 5 and 0.2. Along which eigenvector does the data mostly spread, and what fraction of the variance does it carry?
3. A centred table $\mathbf{X}$ has singular values $\sigma = (4, 3, 1)$. Give the variance explained by the first two components, and the rank of the error $\mathbf{X} - \mathbf{X}_2$.
4. True or false: a real symmetric matrix can have complex eigenvalues. Justify.

- **Determinant** = signed area factor; zero = no inverse. **Rank** = true dimensionality.
- **Outer products**: rank 1 at any size — one profile, one level per person.
- **Eigenvectors** = directions only stretched; $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}$; complex = rotation.
- **Real symmetric** $\Rightarrow$ orthonormal eigenbasis — what makes **PCA** work.
- **SVD** $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$: PCA without forming **C**; best low-rank summary.



Next: differential equations, and why the exponential is the only shape that works.